
Exercises: Advanced Lesson 14

Your Name: Mikey Ferguson

Your Math Class: MATH 7013 - Advanced Mathematical Modeling

1: Using Manipulate[...]:

Part1: Create a dynamic 2-D plot of:

$$e^{-ax} \text{ where } 0 \leq x \leq 1$$

With a continuous slider for a where $0 \leq a \leq 10$

Part2: Note that the plot jumps around as you move the slider. Add a PlotRange option to your plot to fix this problem.

Part: Create a dynamic 3-D plot of:

$$e^{-ax-by} \text{ where } 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 1$$

With a continuous slider for a where $0 \leq a \leq 10$

With a discrete slider for b where $0 \leq b \leq 10$ in steps of 2

```
Manipulate[Plot[E^(-a x), {x, 0, 1}], {a, 0, 10}]
```

```
Manipulate[Plot[E^(-a x), {x, 0, 1}, PlotRange -> All], {a, 0, 10}]
```

```
Manipulate[Plot3D[E^(-a x - b y), {x, 0, 1}, {y, 0, 1}, PlotRange -> All],  
{a, 0, 10}, {b, 0, 10, 2}]
```

2: Using Manipulate[...]:

Create a dynamic algebraic calculator that expands the expression:

$$(x + y)^n$$

With a discrete slider for n where $0 \leq n \leq 10$ in steps of 1

```
Manipulate[Expand[(x + y)^n], {n, 0, 10, 1}]
```

3: For the data set:

dataList = {1,2,3,4,5,6,7}

Part 1: Save the data to your hard disk

Part 2: Display the contents of the disk data file using the FilePrint command

Part 3: Read the disk file into a list and assign the data to symbol: myData

```
Clear[dataList, myData, file];
dataList = {1, 2, 3, 4, 5, 6, 7};
file = FileNameJoin[{$TemporaryDirectory, "dataList.txt"}];
Export[file, dataList, "List"];
file
FilePrint[file]

myData = Import[file, "List"];
myData
{Head[myData], Length[myData], myData === dataList}

4: Convert the text: Mathematica into a list of characters
Characters["Mathematica"]
Length[Characters["Mathematica"]]
```